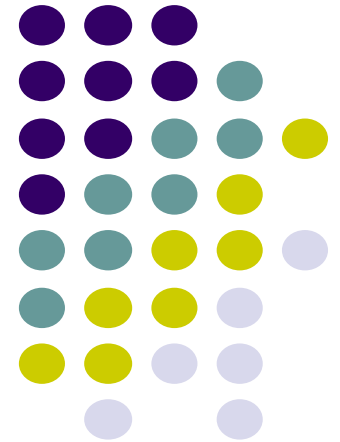


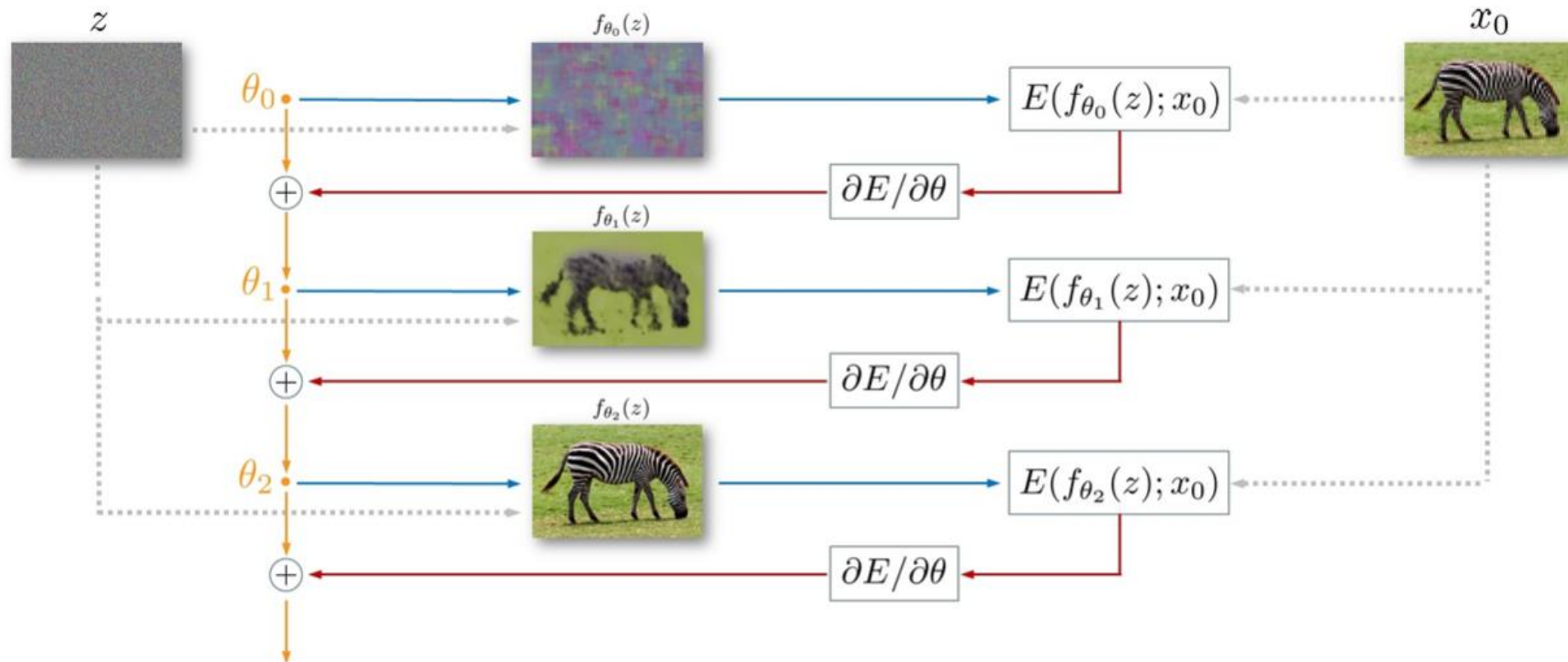
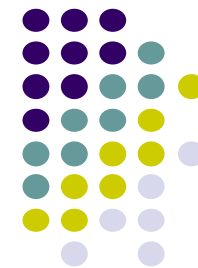
ErSE222: Machine learning in Geoscience

March. 15th, 2026

Tariq Alkhalifah and Omar Saad



Deep Image Prior (DIP)



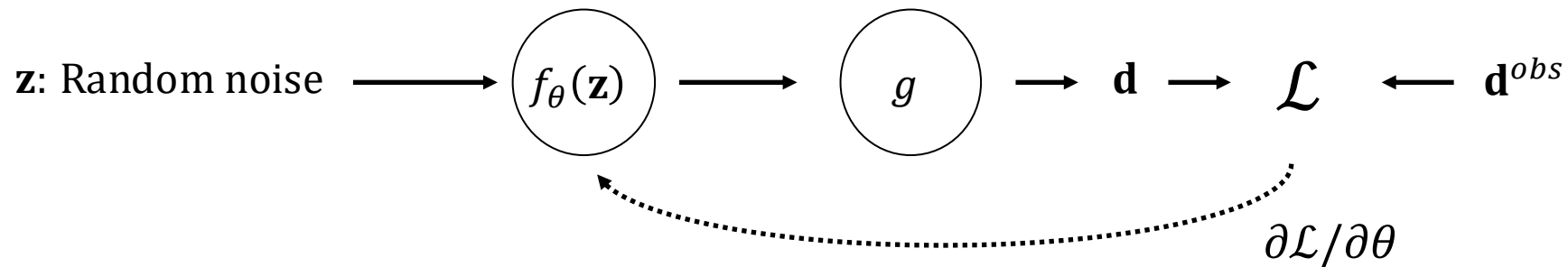
Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 9446-9454).

Deep Image Prior (DIP)

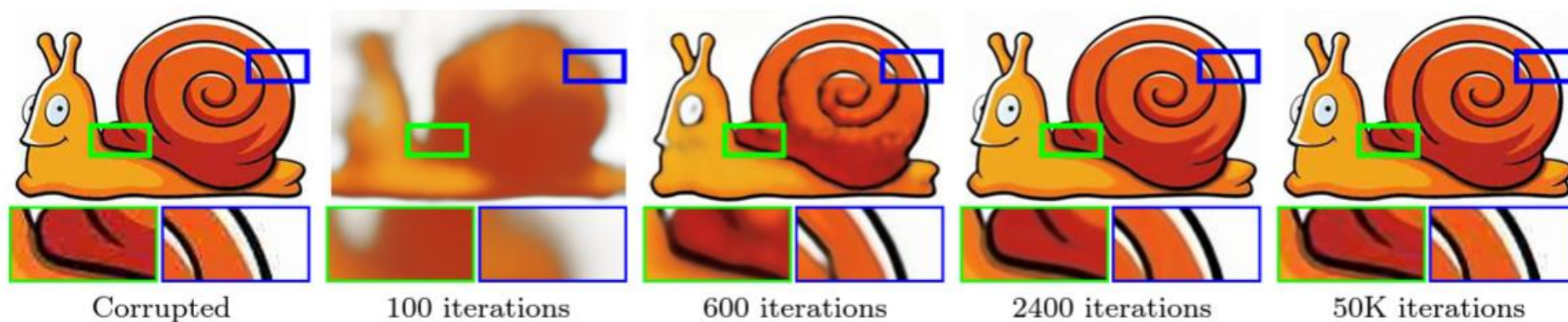
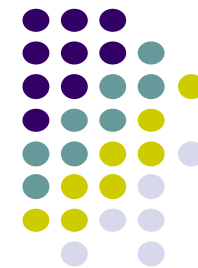


Given a suitable neural network architecture, a set of solutions can be promoted (and the rest rejected)

$$\text{DIP: } J = \|\mathbf{d}^{obs} - g(f_{\theta}(\mathbf{z}))\|_p^p$$



Deep Image Prior (DIP)



Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 9446-9454).

Deep Image Prior (DIP)



Benefits:

- Fully unsupervised, does not require any training sample
- Very easy to implement

Limitations:

- Requires a user to define a suitable network architecture (like for traditional hand-crafter priors)
- Slow convergence

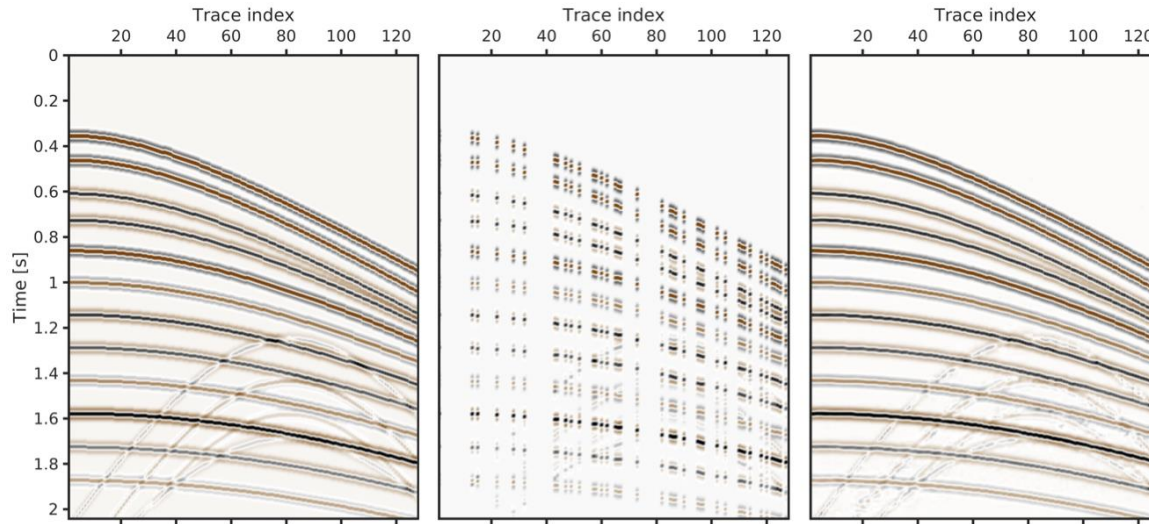


Applications

Deep Image Prior (DIP)



Examples in geophysics



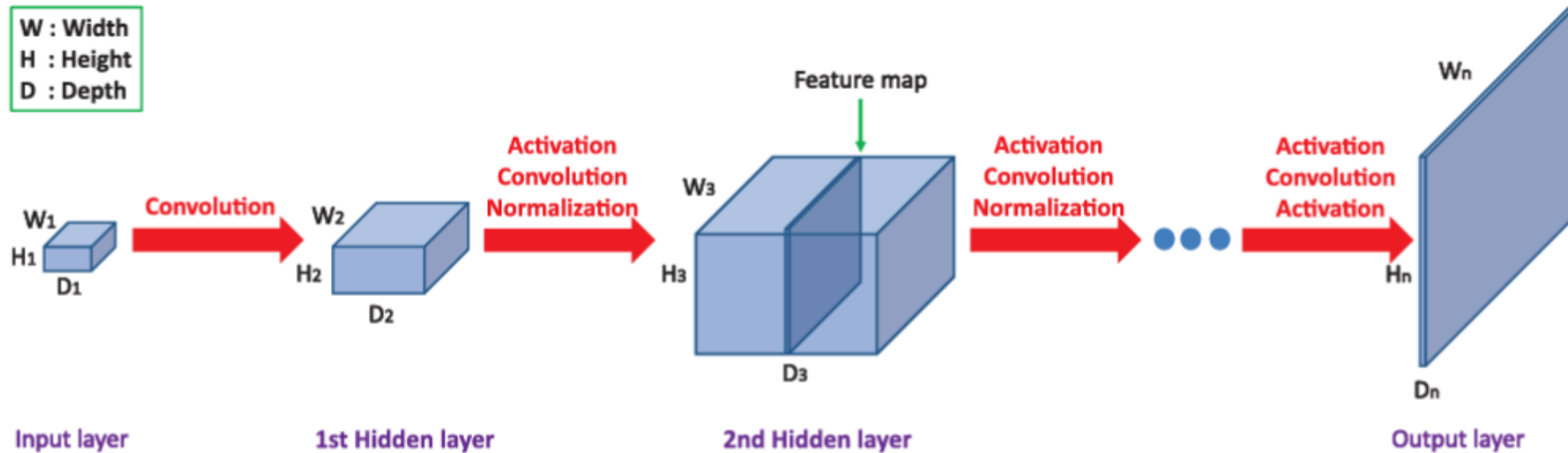
CNNs have natural interpolation capabilities and reject noise

Source: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9310238>

Deep Image Prior (DIP)

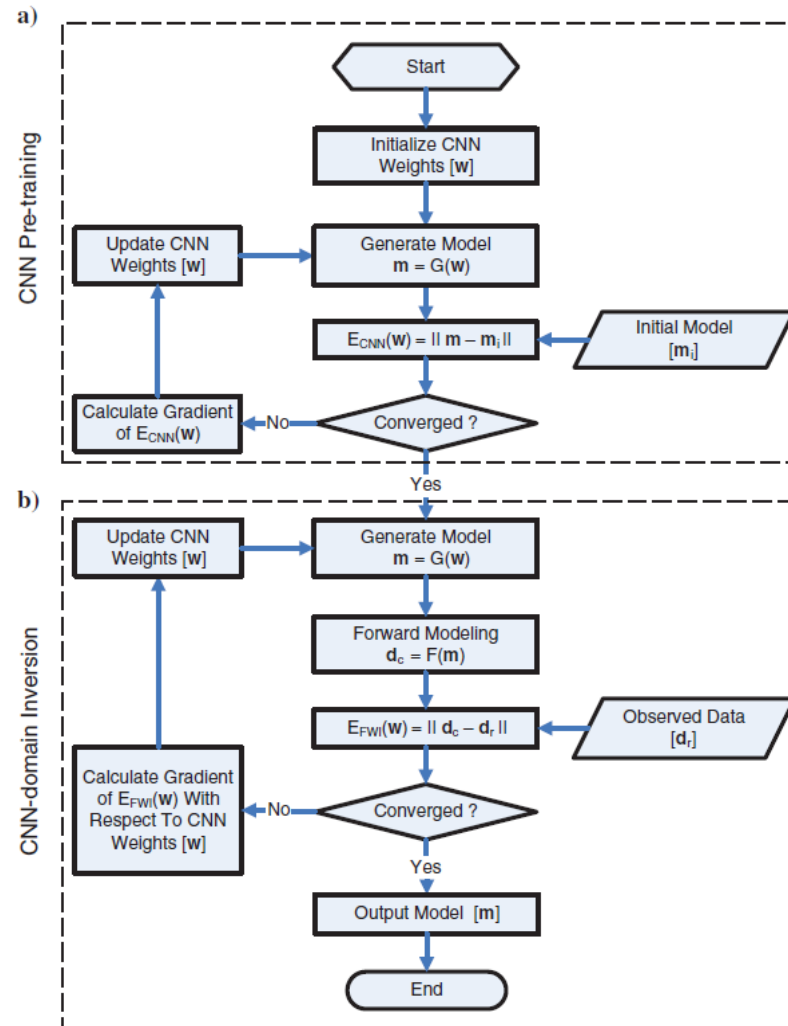


Examples in geophysics

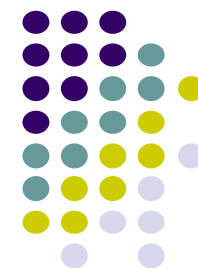


Wu, Y., & McMechan, G. A. (2019). Parametric convolutional neural network-domain full-waveform inversion. *Geophysics*, 84(6), R881-R896.

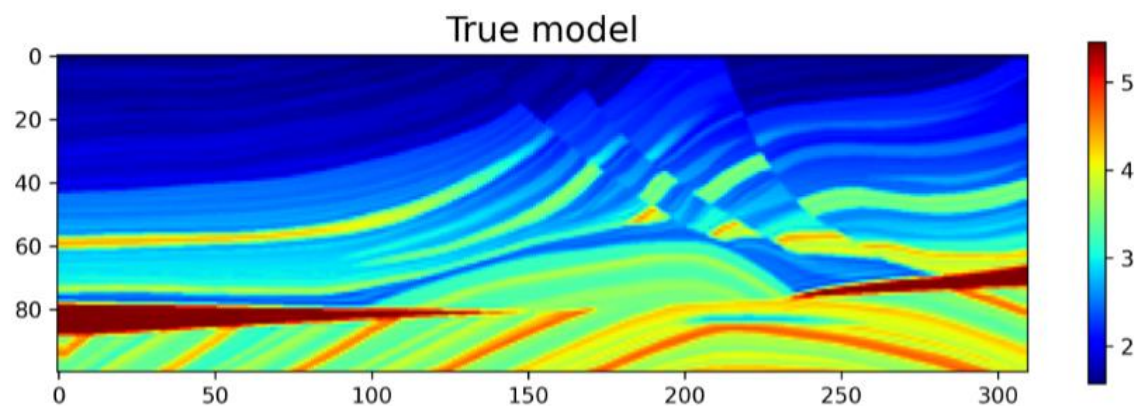
Deep Image Prior (DIP)



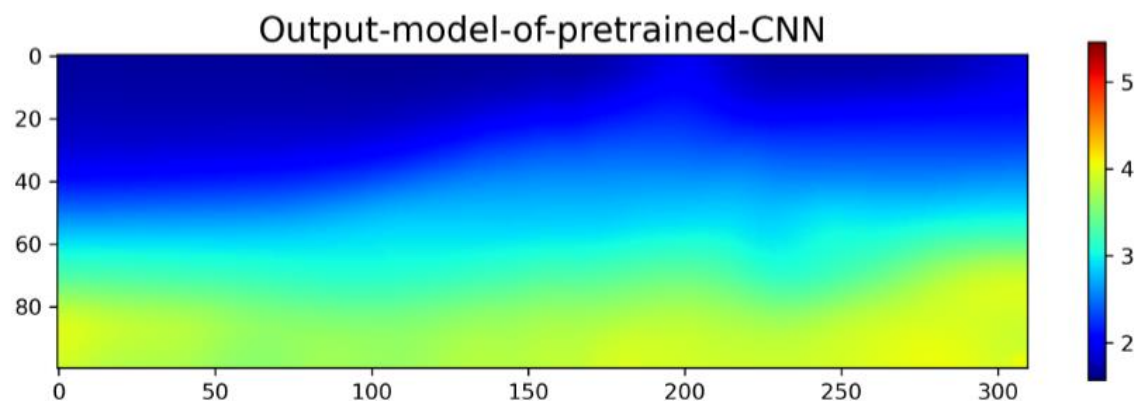
Deep Image Prior (DIP)



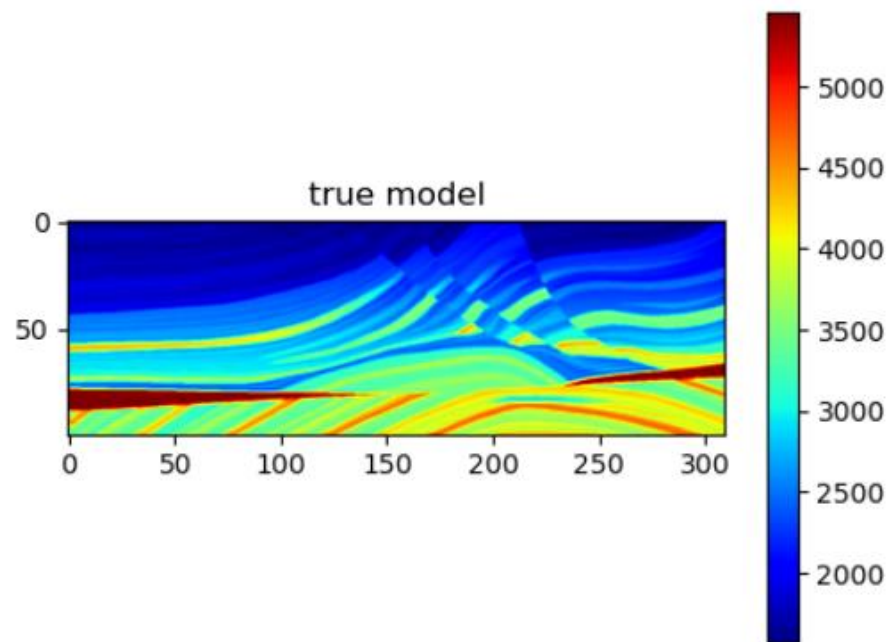
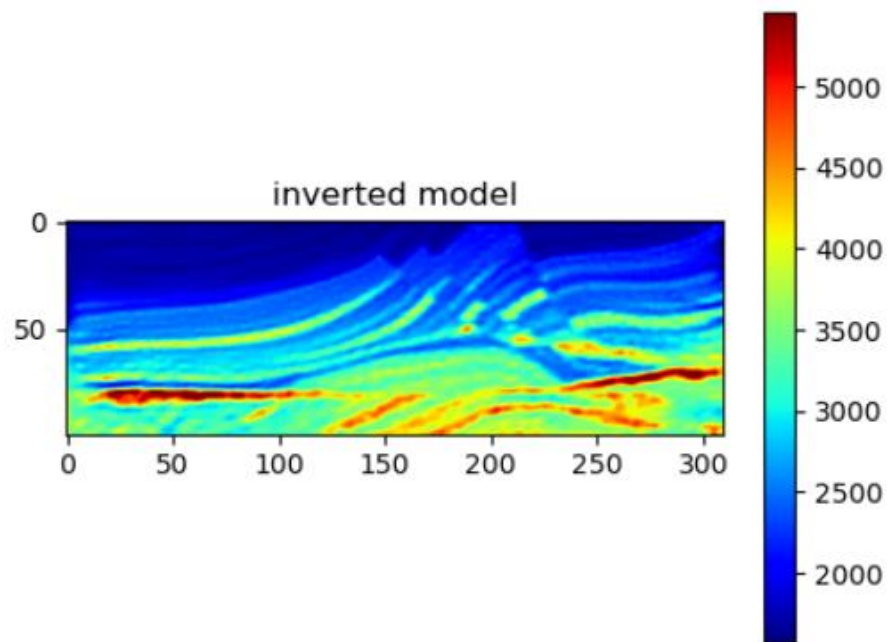
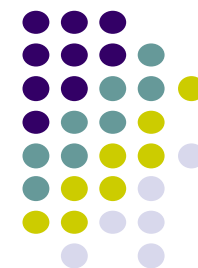
The true model



The initial model



Deep Image Prior (DIP)





Thank You

30 17:50